

Nearshore Rocky Intertidal Overview

Datasets: Rocky Intertidal

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Rocky Intertidal Research Objectives

Rocky shores account for more than 33% of the world's coastal shorelines (Johnson, 1988), and are amongst the dominant coastal habitats in British Columbia (extending along ~50% of the BC coastline). Located between land and sea, where tides ebb and flow, and waves crash, rocky intertidal habitats are characterized by strong environmental gradients and interacting species competing for space and other resources. Many rocky intertidal species are indicators of environmentally driven change to marine ecological communities. As such, they act as sentinels to the biological costs of climate change, from individuals, to population and community level effects (Denny, 1987; Helmuth et. al., 2002, Harley, 2011).

Macrophytes in the rocky intertidal are important components of the rocky intertidal community. One cultural indicator species, the red alga *Pyropia abbotiae*, provides a food source for coastal British Columbia indigenous communities. Generations of harvesters have relied on this seaweed for food and trade (Turner 2003). However, Heiltsuk, Wuikinuxv, Kitasoo, Haida, and Metlakatla communities all reported declines in their annual seaweed harvests in 2016 – the worst harvest in at least 100 years (Mike Reid, Heiltsuk Nation). This provides a focal point for rocky intertidal research on indicator species, and the climate and ocean-based drivers of their dynamics.

The Hakai Institute Rocky intertidal program aims to evaluate the trends and key drivers of change within the rocky intertidal ecosystem on Calvert Island, BC. We are dedicated to documenting annual trends in biodiversity and annual and seasonal trends of indicator species to achieve focal research objectives:

Research Objectives

- Track seasonal and annual change of indicator species productivity and biodiversity
- Determine the drivers of change in the rocky intertidal ecosystems
- Assess resilience of rocky intertidal species to disturbance, stress and disease
- Understand connectivity between rocky intertidal habitats and other coastal ecosystems

Active Research Questions

1. What are the seasonal and interannual drivers of variability in rocky intertidal indicator species (mussels, *Fucus*, *Phyllospadix*, barnacles)?
2. Does rocky intertidal habitat complexity drive biodiversity of motile invertebrates?
3. Are rocky intertidal habitats on Calvert Island resilient to disturbance, stress and disease?
4. What are the costs of asteroid idiopathic wasting syndrome for intertidal sea stars?
5. What drives change in *Pyropia abbotiae* production and quality?

Rocky Intertidal Datasets

Datasets & Parameters	Years
Pisaster wasting disease: abundance , size , wasting disease status, <i>Katharina tunicata</i> (Katy chiton) abundance & size	Seasonal: 2016-2025
Barnacle diversity: abundance and diversity of Balanus glandula , <i>Semibalanus cariosus</i> , and Chthamalus spp.	Seasonal: 2016-2025
Mussel abundance: abundance , length , bed depth , growth of <i>Mytilus californianus</i>	Seasonal: 2016-2025
Fucus cover & biomass : <i>Fucus distichus</i> percent cover and diversity of other algae	Seasonal: 2016-2025
Phyllospadix cover & biomass: percent cover, bed thickness, shoot density, flowering shoot density, epiphytes, canopy height	Seasonal: 2016-2025
Motile invertebrate diversity: abundance, diversity, patchiness, substrate associations in barnacle, mussel, <i>Fucus</i> and <i>Phyllospadix</i> surveys (e.g. limpets , snails , crabs)	Seasonal: 2016-2025
Pyropia abbotiae productivity: percent cover , abundance , growth , health status	Seasonal: 2017-2019
Pyropia abbotiae collections: biomass, stable isotopes/C:N, pigment concentration, morphology, barnacle collections for sporophyte stage, timing of when reproduction occurs	Seasonal: 2017-2019
Pyropia community diversity: algal diversity, & percent cover, grazer abundance & diversity	Seasonal: 2017-2018
Habitat zonation mapping: orthomosaics of mussels, fucus, barnacles, <i>Pyropia</i> and <i>phyllospadix</i> (UAV)	Seasonal: 2018
Macrophyte isotopes: C:N, d13C, d15N for <i>Pyropia</i> spp., <i>Fucus</i> , <i>Ulva</i>	Seasonal: 2016-2019
Sea star wasting disease: abundance, size, wasting disease status, epidermal microbial diversity, genomic diversity	Seasonal: 2019-2025

For further details on field and lab protocols, see "[Hakai Rocky Intertidal Protocol.pdf](#)".

Data Packages

Priority data package releases

1. Seastar Populations & Wasting Disease [2017-2023](#)
2. Barnacle Abundance & Spatial Coverage [2019-2023](#)
3. Mussel Bed Dynamics [2016-2023](#)
4. Fucus Bed Dynamics [2016-2023](#)
5. Phyllospadix Bed Community Dynamics [2017-2023](#)
6. Mobile Invertebrate Diversity [2016-2023](#)
7. *Pyropia abbotiae* 2017-2019 (private, available upon request)
8. Rocky Intertidal Isotopes (lead: Froese): [Nearshore macrophyte stable isotopes](#)

2023 Activities

Core activities

- Seasonal *Pisaster ochraceus*, *Katharina tunicata*, barnacle, mussel, *Fucus spp.*, motile invertebrates, *Phyllospadix spp.* and sea star wasting disease surveys
 - sites: North Beach, Fifth Beach, West Beach
 - summer months (June, Sept), winter months (Feb, Nov).
 - provides core LTER & MarineGEO monitoring to document changes in rocky intertidal habitat diversity, productivity, and the drivers of change
 - paired aerial surveys track changes in extent, coupled with population and community dynamics of focal species, their diversity, disease prevalence and ecological functions
- Mussel growth experiments
 - Annual intervals in August
- Biodiversity survey
 - Biannual (July or August) on Calvert PISCO site and Duck Island PISCO and BC parks site
- Seasonal intertidal and subtidal sea star wasting disease surveys
 - sites: North Beach (RI and Nereo), Fifth Beach, West Beach (RI and macro), Kelpie, Mercury, Odlum, Thrasher, Kwakshua, Kwakume, Goodlad, Mustang, Meay, Second (Nereo), Triquet, Choked Pass and Pruth Bay
 - May, August and October

Correlates

- Paired UAV mapping with surveys (Hakai Geospatial)

- Change detection in extent of mussels and *Phyllospadix* spp.
- Nearshore environmental parameters
 - Seasonal temperature loggers
 - YSI measurements: temperature, SPC, salinity, pH (single reading at surface (0m))
 - Hyper-nearshore water collections at surface (0m) to provide chlorophyll, nutrients, do13C samples

Collaborations & Network Initiatives

- Collaboration with Dr. Chris Harley (UBC) and Dr. Alyssa Gehman (UBC)
 - Sea star wasting disease across nearshore habitats (Kwakwaka'wakw)
 - Seastar effects on species richness
 - Nucella reproduction, distribution and competition
 - Parasite diversity in the hermit crab *Pagurus hirsutiusculus*
- Collaboration with Dr. Patrick Martone (UBC)
 - Annual seaweed survey
 - Coralline algae experiments
- MarineGEO
 - global network initiatives: assessment of field method standards, habitat-specific biodiversity baselines for global comparative analyses

2016-2022: Activities Completed

- Seasonal *Pisaster*, barnacle, mussel, *Fucus* and motile invert surveys 2016-2022 (North Beach, West Beach and Fifth Beach): 3 summer, 1-2 winter time seasonal time points
- Mussel growth experiments 2017-2022: six month intervals (May, October)
- Recruitment time series (Dr. Matt Whalen, Post Doc)
 - Seasonal invertebrate recruitment assays in rocky intertidal (PISCO site, West Beach and Fifth Beach)
- Seasonal *Pyropia* collections and surveys. Project complete 2020
 - Spring months (Mar-June)
 - 2017: Seasonal (May, June, August), 2018: Seasonal (Feb-July), 2018-2019 Annual survey with Coastal Guardian Watchmen (Apr/May)
 - Macrophyte isotopes 2016-2018

- Seasonal biomass, pigment concentration, morphology, barnacle collections for *Pyropia* sporophyte stage. Biweekly in the spring.
- *Pyropia conchocelis* surveys (Dr. Jenn Clark, Post doc)
 - Seasonal phenology, timing and location of *Pyropia* conchocelis
 - Quantifying wave force (Foggy Cove, Little Wolf and Nalau Passage)
 - Testing restoration methods of *Pyropia* (2019-2020)
- Collaboration with PISCO and Dr. Luran Schielbelhut (UC Davis) and Dr. Michael Dawson (UC Merced) - Biodiversity survey
 - Sea star survey at Calvert PISCO site (2018)
 - Collaboration with Coastwide survey of *Pisaster* genetics (2018)
 - Annual 2017 at Calvert PISCO site and Duck Island (July)
- Collaboration with Dr. Laura Parfrey (UBC)
 - 2019 - *Fucus* microbiome sampling & transplant experiments
- Collaboration with PISCO
 - Coastwide rocky intertidal biodiversity surveys
- Coastal Guardian Watchmen Network
 - Development of methods for standardized annual survey of *Pyropia* for Watchmen
 - Training (Watchmen meeting in May 2018)
 - Annual survey (Apr/May 2018)
- Rocky Intertidal Bioblitz 2017
 - First Calvert BioBlitz (led by Dr. Matt Whalen, postdoc)
- Nearshore environmental parameters 2017-2018
 - Seasonal temp loggers (tidbit), YSI and hyper-nearshore water sampling
- Collaboration with the Harley lab (UBC)
 - Dr. Alyssa Gehman (UBC) and Dr. Chris Neufeld (Bamfield marine sciences centre), mussel biodiversity surveys (2017)
 - Kwakshua biodiversity survey (2015-2018)
 - *Nucella* reproduction, distribution and competition (2017)
 - Dr. Alyssa Gehman (postdoc) mussel-cyanobacterial interactions (2017-2018)
 - *Pisaster* abundance and SSWD incidence (2014-2018)
 - *Mytilus trossulus* collection for Miling Li (UBC) to [measure Pb isotopes](#) (2018)

- Collaboration with Dr. Carrie Keogh (Emory) and Dr. Alyssa Gehman (UBC)
 - Variation in infection prevalence in hermit crabs (2018)
- Collaboration with the Martone lab (UBC)
 - Annual seaweed surveys (2010-2017)
- Collaboration with the Parfrey lab (UBC)
 - Katy Davis (PhD student) fucus microbiome transplants (2017)
 - Dr. Alyssa Gehman (postdoc) mussel shell microbial communities (2017)

Publications & Reports (Published & Submitted)

Heady, W.N., R. Beas-Luna, M.N Dawson, N. Eddy, K. Elsmore, F. T. Francis, T. Frierson, A.-L. M. Gehman, T. Gotthardt, S.A. Gravem, J. Grebel, S. L. Hamilton, L. Hannah, C.D. Harvell, J. Hodin, I. Kelmartin, C. Krenz, L. Lee, J. Lorda, D. Lowry, S. Mastrup, E. Meyer, P. T. Raimondi, S. S. Rumrill, V. R. Saccomanno, L. M. Schiebelhut, C. Siddon. 2022. Roadmap to recovery for the sunflower sea star (*Pycnopodia helianthoides*) along the west coast of North America. The Nature Conservancy, Sacramento, CA, US. 44 pages. <https://bit.ly/SunflowerSeaStar>

Davis, K. M., F. Mazel, and L. W. Parfrey. 2021. The microbiota of intertidal macroalgae *Fucus distichus* is site-specific and resistant to change following transplant. *Environmental Microbiology*.

Lemay, M. A., K. M. Davis, P. T. Martone, and L. W. Parfrey. 2021. Kelp-associated microbiota are structured by host anatomy. *Journal of Phycology*.

Gravem, S.A., W.N. Heady, V.R. Saccomanno, K.F. Alvstad, A.M. Gehman, T.N. Frierson, S.L. Hamilton. 2020. *Pycnopodia helianthoides*. IUCN Red List of Threatened Species 2020, e.T178290276A178341498.

Gehman, A.M., C. Keogh, D.A. Satterfield, A.K. McKay, S. Budischak. 2019. To improve ecological understanding, collect infection data. *Ecosphere*, 10.1002/ecs2.2770

*Li, M., D. Weis, K. E. Smith, A. E. Shiel, W. D. Smith, B. P. V. Hunt, A. Torchinsky, and E. A. Pakhomov. 2019. Assessing lead sources in fishes of the northeast Pacific Ocean. *Biochemical Pharmacology* 29:100234. [pdf](#)

Lemay, M., Martone, P.T., Keeling, P., Burt, J., Krumhansl, K., Sanders, R., and L. Parfrey. (2018) Sympatric kelp species share a large portion of their surface bacterial communities. *Environ. Microbiol.* 20(2): 658-670 [pdf](#)

Lemay, M., Martone, P.T., Hind, K., Lindstrom, S., and L.W. Parfrey. (2018) Alternate life history phases of a marine macroalgal genus have distinct microbial surface communities. *Mol. Ecol.* 00:1–14. [pdf](#)

Hind, K., Miller, K.A., Young, M., Jensen, C., Gabrielson, P.W., and P.T. Martone. (2015) Resolving cryptic species of *Bossiella* (Corallinales, Rhodophyta) using contemporary and historical DNA. *Am. J. Bot.* 102(11): 1-19 [pdf](#)

Starko, S. and P.T. Martone. (2016) An empirical test of 'universal' biomass scaling relationships in kelps: evidence of convergence with seed plants. *New Phytol.* 212: 719-729 [pdf](#) [cover](#) [blogpost](#) [Science press](#)

Lindstrom, S. (2017) An undescribed species of putative Japanese *Pyropia* first appeared on the central coast of British Columbia, Canada, in 2015. *Mar. Poll. Bull.* [pdf](#)

*Collections by the rocky intertidal program, staff in acknowledgments

Technical Documents

Sadlier-Brown, G and Gehman, A. **2018**. Hakai Rocky Intertidal Protocols 2018. Hakai Technical Document.

Clark J., Liggan L., Martone P., Lindstrom, S. Hessing-Lewis, M., 2019. Annual *Pyropia abbotiae* harvest surveys from 2017-2019 collected near Calvert Island, British Columbia, Canada. Hakai Institute.

Presentations & Outreach

Western Society of Naturalists (November 2019)

- Gehman, A., O. Pontier, G. Sadlier-Brown, T. Froese
- Oral Presentation: Variation in species abundance following sea star wasting disease outbreak in Central British Columbia, Canada.

Society of Integrative and Comparative Biology (2019)

- Gehman, A., O. Schaefer.
- Oral presentation: Drivers of variation in host and parasite performance

University of Auckland (Sept, 2018)

- Harley, C.D.G.
- Invited talk: *Species interactions, environmental change, and biodiversity on rocky shores.*

Ecological Society of America (Aug, 2018)

- Gehman, A., O. Schaeffer and C. Harley.
- Oral Presentation: *Drivers of variation in host and parasite performance*

Phycological Society of America (July, 2018)

- Clark, J.S.
- Oral Presentation: *The legacy of the blob: ongoing effects of the Pacific warm water anomaly on the traditionally harvested alga, *Pyropia abbotiae**
- Liggan, L.
- Oral Presentation: *Documenting population dynamics of a traditionally harvested seaweed: development of long-term monitoring surveys.*

International Echinoderm Conference (June, 2018)

- Gehman, A. and O. Pontier.
- Oral Presentation: *Progression of sea star wasting disease outbreak in central BC, Canada.*

World Conference on Marine Biodiversity (May, 2018)

- M.A. Whalen, M. Hessing-Lewis, M.A. Lemay, G. Paulay, J.E. Duffy, and the Hakai BioBlitz Team.
- Oral Presentation: *The 2017 Hakai-MarineGeo BioBlitz: a new gold standard approach to inventorying and understanding marine biodiversity.*

Wuikinuxv council (May 2018)

- Clark, J.
- Oral Presentation: *Coastwide Seaweed (Pyropia) Observatory Program*

Heiltsuk Integrated Resource Management Department (HIRMD) (Oct 2018)

- Clark, J.
- Oral Presentation: *Coastwide Seaweed (Pyropia) Observatory Program*

Heiltsuk Community Presentation (Oct 2018)

- Clark, J.
- Oral Presentation: *Coastwide Seaweed (Pyropia) Observatory Program*

Heiltsuk Classroom Outreach (Oct 2018)

- Clark, J.
- Classroom presentation with elementary students about seaweeds on Central Coast

Student workshop with Heiltsuk, Nuxalk and Kitsoo (2018)

- Clark, J. and Liggan, L.
- seaweed walk and seaweed presses

Western Society of Naturalists (Nov. 2017)

- Harley, C.D.G., A.M. Gehman, & C. Neufeld.
- Oral Presentation: *Species richness is not maximized by keystone predation per se, but by variation in keystone predator effects.*

Bodega Marine Laboratory Seminar (2017)

- Harley, C.D.G.
- Invited Seminar: *The importance of species interactions in an era of environmental change.*

University of British Columbia Biodiversity seminar (2017)

- Harley, C.D.G.
- Invited seminar: *Abiotic stressors and species interactions in an era of environmental change*

University of Victoria Seminar series (2017)

- Harley, C.D.G.
- Invited seminar: *When do species interactions dictate the ecological responses to environmental change?*

Training

Coastal Guardian Watchman Annual Pyropia Survey (May, 2018)

- Clark, J., Liggan, L. and Hessing-Lewis, M.

Rocky Intertidal Research Sites Map



Detailed UAV maps of survey design for the biodiversity and dynamics can be found [Fifth Beach](#), [North Beach](#) and [West Beach](#) (source: Keith Holmes).

Literature Cited

- Harley, C. D. G. 2011. *Climate Change, Keystone Predation, and Biodiversity Loss*. *Science* 334:1124–1127.
- Harley, C. D., Anderson, K. M., Demes, K. W., Jorve, J. P., Kordas, R. L., Coyle, T. A., & Graham, M. H. (2012). *Effects of climate change on global seaweed communities*. *Journal of Phycology*, 48(5), 1064-1078.

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- van der Heijden, L. H., and N. A. Kamenos. 2015. *Reviews and syntheses: Calculating the global contribution of coralline algae to total carbon burial. Biogeosciences* 12:6429–6441.
- Thompson, R. C., T. P. Crowe, and S. J. Hawkins. "Rocky intertidal communities: past environmental changes, present status and predictions for the next 25 years." *Environmental Conservation* 29.2 (2002): 168-191.